

Q1. ID numbers, full names, and addresses are duplicated.

Q2. Only one file was updated so the data no longer matches.

Q3. Since the files aren't synced, updates are only applied to one file even if they originally had the same information.

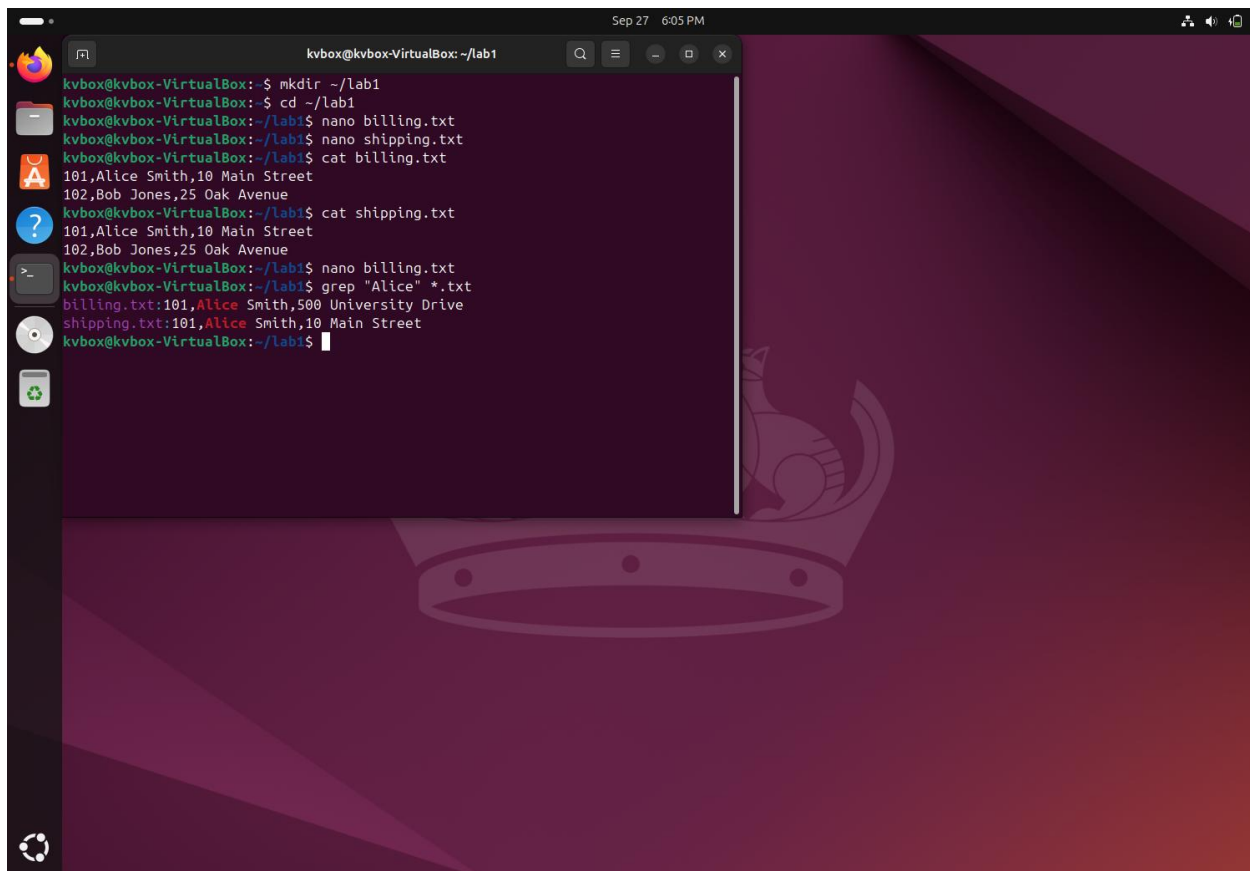
Q4. The database is the actual "container" of data, and the DBMS is the system used to manage the database.

Q5. Only one record had to be updated.

Q6. This is less likely to have inconsistent information because the data is stored in a single location so one change is applied everywhere.

Q7. This provided the table structure with a breakdown of the columns, which column acts as the primary key, the data types, and if NULL values are allowed.

Q8. This is considered metadata because it's data about data.



```
kvbox@kvbox-VirtualBox: ~/lab1
kvbox@kvbox-VirtualBox:~$ mkdir ~/lab1
kvbox@kvbox-VirtualBox:~$ cd ~/lab1
kvbox@kvbox-VirtualBox:~/lab1$ nano billing.txt
kvbox@kvbox-VirtualBox:~/lab1$ nano shipping.txt
kvbox@kvbox-VirtualBox:~/lab1$ cat billing.txt
101,Alice Smith,10 Main Street
102,Bob Jones,25 Oak Avenue
kvbox@kvbox-VirtualBox:~/lab1$ cat shipping.txt
101,Alice Smith,10 Main Street
102,Bob Jones,25 Oak Avenue
kvbox@kvbox-VirtualBox:~/lab1$ nano billing.txt
kvbox@kvbox-VirtualBox:~/lab1$ grep "Alice" *.txt
billing.txt:101,Alice Smith,500 University Drive
shipping.txt:101,Alice Smith,10 Main Street
kvbox@kvbox-VirtualBox:~/lab1$
```

```
Sep 27 6:42 PM
kvbox@kvbox-VirtualBox: ~/lab1

Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO customer VALUES
-> (101, 'Alice Smith', '10 Main Street'),
-> (102, 'Bob Jones', '25 Oak Avenue');
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM customer;
+-----+-----+-----+
| customer_id | name      | address      |
+-----+-----+-----+
| 101 | Alice Smith | 10 Main Street |
| 102 | Bob Jones  | 25 Oak Avenue  |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> UPDATE customer
-> SET address = '500 University Drive'
-> WHERE customer_id = 101;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM customer;
+-----+-----+-----+
| customer_id | name      | address      |
+-----+-----+-----+
| 101 | Alice Smith | 500 University Drive |
| 102 | Bob Jones  | 25 Oak Avenue  |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> DESCRIBE customer;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| customer_id | int       | NO   | PRI | NULL    |       |
| name        | varchar(100) | YES  |     | NULL    |       |
| address     | varchar(150) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

mysql>
```